Theoretical Spectroscopy via Many-Body Perturbation Theory

April 5, 2026

Outline

1. Introduction
2. Methods
3. Future directions

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Motivation

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- Spectroscopy helps discover these materials
  - Theoretical spectroscopy facilitates the interpretation of experiments

Spectral functions of the Uniform Electron Gas (UEG)

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Approach	**Pros**	**Cons**
EOM-CC	Captures satellite features	Expensive
Perturbative expansion of Dyson equation	Computationally efficient	Does not capture satellite features
Cumulant expansion	Computationally efficient and captures satellite features	May incorrectly predict the location of satellite features

Cumulant expansion

For a quasiparticle indexed by p,

$$G^R_{pp}(t) = G^R_{0,pp}(t) e^{C^R_{pp}(t)}, \tag{1}$$

where

$$C^R_{pp}(t) = -\frac{1}{\pi} \int d\omega \frac{\mathrm{Im}\Sigma^R_{pp}(\omega)}{\omega^2} \left(e^{-i\omega t} + i\omega t - 1 \right). \tag{2}$$

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I will explore three different approximations for $\Sigma(\omega)$: GW, PT2, and BWs2.

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GW+C

$$\Sigma^{GW}(\omega) = G^{HF}(\omega)^{-1} - G(\omega)^{-1} \tag{3}$$

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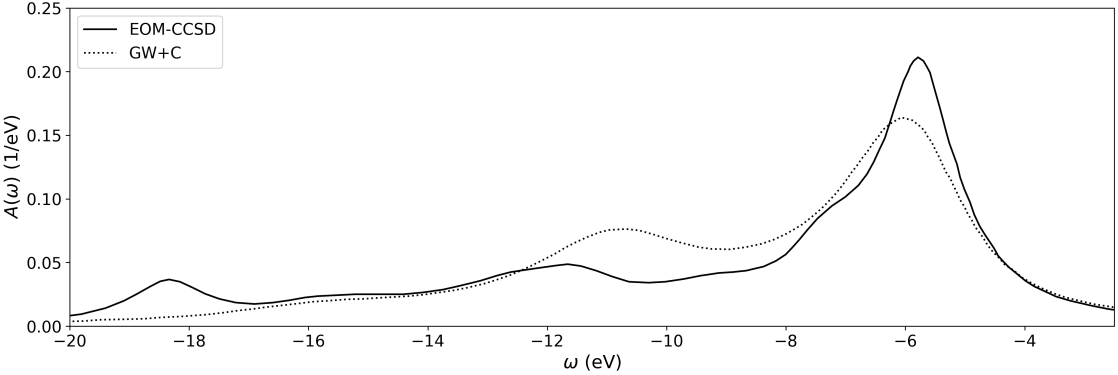

PT2+C

$$\Sigma^{(2)}(\omega) = G^{HF}(\omega)^{-1} - G(\omega)^{-1} \tag{4}$$

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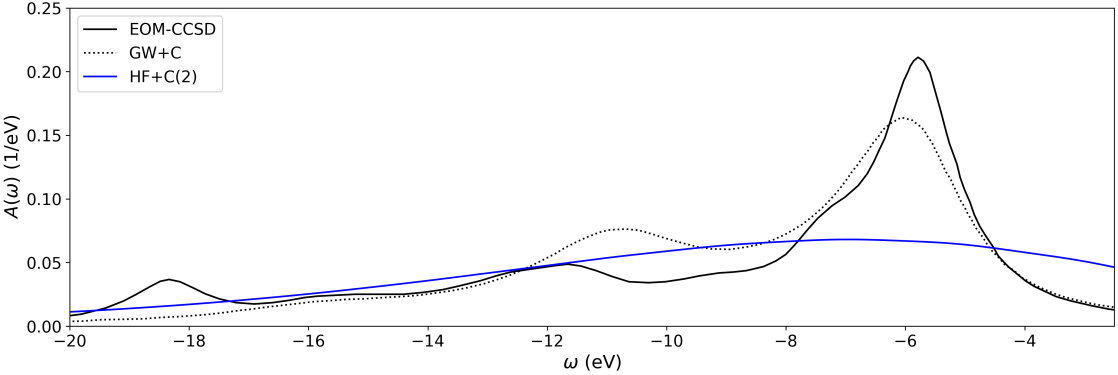

BWs2+C

$$\Sigma^{BWs2}(\omega) = G^{BWs2}(\omega)^{-1} - G(\omega)^{-1} \tag{5}$$

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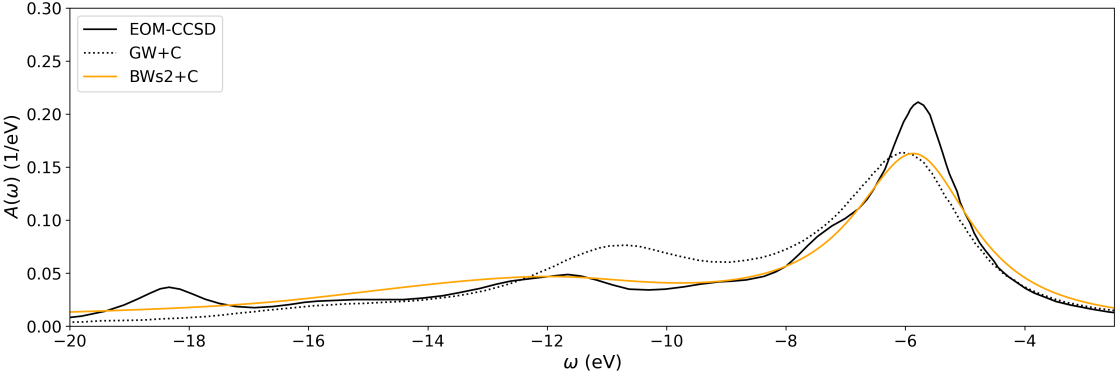

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Self-consistent cumulant

1. Solve for the cumulant via

$$C^R_{pp}(t) = -\frac{1}{\pi}\int d\omega \frac{\mathrm{Im}\Sigma^R_{pp}(\omega)}{\omega^2}\left(e^{-i\omega t} + i\omega t - 1\right). \tag{6}$$

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and plug into equation 6.

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4. Iterate until self-consistency

Memory kernel formulation

The GQME describes the time evolution of a correlation function $\mathcal{C}(t)$ as

$$\dot{\mathcal{C}}(t) = \mathcal{C}(t)\Omega_1 - \int_0^t d\tau\, \mathcal{C}(t-\tau)\mathcal{K}_1(\tau) + D(t), \tag{9}$$

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Quantity	**Expansion Parameter**	**Decay**	**Correlation Strength**
$\mathcal{K}_1(t)$	t	Fast	Strong
$\Sigma^R(t)$	V	Slow	Weak to moderate

Questions?